

Pharmaniaga Doxycycline Capsule 100mg

DESCRIPTION

Maroon/maroon, size '2', hard gelatin capsule with Pharmaniaga icon and '100' markings.

COMPOSITION

Each capsule contains Doxycycline HCl equivalent to Doxycycline 100mg.

PHARMACODYNAMICS

Mechanism of action:

Doxycycline is primarily bacteriostatic and is believed to exert its antimicrobial effect by the inhibition of protein synthesis.

Doxycycline is active against a wide range of Gram-positive and Gram-negative bacteria and certain other micro-organisms.

PHARMACOKINETICS

Absorption

Tetracyclines are readily absorbed and are bound to plasma proteins in varying degrees.

Doxycycline is virtually completely absorbed after oral administration. Studies reported to date indicate that the absorption of doxycycline, unlike certain other tetracyclines, is not notably influenced by the ingestion of food or milk.

Following a 200mg dose, normal adult volunteers averaged peak serum levels of 2.6 micrograms/ml of doxycycline at 2 hours decreasing to 1.45 micrograms/ml at 24 hours.

Doxycycline has a high degree of lipid solubility and a low affinity for calcium.

Biotransformation

It is highly stable in normal human serum. Doxycycline will not degrade into an epianhydro form. No significant metabolism occurs.

Elimination

Doxycycline is concentrated in bile by the liver and excreted largely unchanged in urine and faeces.

INDICATIONS

Treatment:

Doxycycline is indicated for treatment of the following infections:

- Rocky Mountain spotted fever, typhus fever and the typhus group;
- Q fever, rickettsial pox and tick fevers caused by *Rickettsiae*;
- Respiratory infections caused by *Mycoplasma pneumoniae*;
- Psittacosis caused by *Chlamydia psittaci*;
- Lymphogranuloma venereum, caused by *Chlamydia trachomatis*
- Uncomplicated urethral, endocervical or rectal infections in adults caused by *Chlamydia trachomatis*;
- Trachoma caused by *Chlamydia trachomatis* although the infectious agent is not always eliminated, as judged by immunofluorescence;

- Inclusion conjunctivitis caused by *Chlamydia trachomatis* may be treated with oral doxycycline alone or with a combination of topical agents;
- Granuloma inguinale (donovanosis) caused by *Calymmatobacterium granulomatis*;
- Early (Stage 1) Lyme disease caused by *Borrelia burgdorferi*;
- Louse-borne relapsing fever caused by *Borrelia recurrentis*;
- Tick-borne relapsing fever caused by *Borrelia duttonii*;
- Non-gonococcal urethritis (NGU) caused by *Ureaplasma urealyticum* (*T-Mycoplasma*).

Doxycycline is also indicated for the treatment of infections caused by the following gram-negative microorganisms:

- Acinetobacter* species;
- Bacteroides* species;
- Fusobacterium* species;
- Brucellosis caused by *Brucella* species (in conjunction with streptomycin);
- Plague caused by *Yersinia pestis*;
- Tularemia caused by *Francisella tularensis*;
- Bartonellosis caused by *Bartonella bacilliformis*;
- Campylobacter fetus*.

Because many strains of the following groups of microorganisms have been shown to be resistant to tetracyclines, culture and susceptibility testing are recommended.

Doxycycline is indicated for treatment of infections caused by the following gram-negative microorganisms, when bacteriologic testing indicates appropriate susceptibility to the drug:

- Shigella* species;
- Uncomplicated gonorrhea caused by *Neisseria gonorrhoeae*;
- Respiratory infections caused by *Haemophilus influenzae*;
- Respiratory and urinary infections caused by *Klebsiella* species;
- Escherichia coli*;
- Enterobacter aerogenes*;
- Moraxella catarrhalis*.

Doxycycline is indicated for treatment of infections caused by the following gram-positive microorganisms when bacteriologic testing indicates appropriate susceptibility to the drug:

- Streptococcus* species: A certain percentage of strains of *Streptococcus pyogenes* and *Streptococcus faecalis* have been found to be resistant to tetracycline drugs. Tetracyclines should not be used for streptococcal infections unless the organism has been demonstrated to be sensitive.
- For upper respiratory infections due to group A beta-hemolytic streptococci, penicillin is the usual drug of choice, including prophylaxis of rheumatic fever. This includes:
-Upper respiratory tract infections caused by *Streptococcus pneumoniae*;

Respiratory, skin and soft-tissue infections caused by *Staphylococcus aureus*. Tetracyclines are not the drug of choice in the treatment of staphylococcal infections.

When penicillin is contraindicated, doxycycline is an alternative drug in the treatment of:

- Actinomycosis caused by *Actinomyces* species;
- Infections caused by *Clostridium* species;
- Syphilis caused by *Treponema pallidum* and yaws caused by *Treponema pertenue*;
- Listeriosis caused by *Listeria monocytogenes*;
- Vincent’s infection (acute necrotizing ulcerative gingivitis) caused by *Leptotrichia buccalis* (formerly, *Fusobacterium fusiform*).

Adjunctive treatment

In acute intestinal amebiasis, doxycycline may be a useful adjunct to amebicides.

In severe acne caused by acne vulgaris, doxycycline may be useful adjunctive therapy.

Treatment and Prophylaxis

Doxycycline is indicated for the prophylaxis and treatment of the following infections:

- Malaria caused by *Plasmodium falciparum* (in areas with chloroquine-resistant *P. falciparum*).
- Leptospirosis caused by genus *Leptospira*.
- Cholera caused by *Vibrio cholerae*.

Prophylaxis

Doxycycline is indicated as prophylaxis in the following conditions:

- Scrub typhus caused by Rickettsia *tsutsugamushi*;
- Traveler's diarrhea caused by enterotoxigenic *Escherichia coli*.

CONTRAINDICATIONS

- Hypersensitivity to the active substance, any of the tetracyclines or to any of the excipients.
- Pregnancy:** Doxycycline is contraindicated in pregnancy. It appears that the risks associated with the use of tetracyclines during pregnancy are predominantly due to effects on teeth and skeletal development.
- Nursing mothers:** Tetracyclines are excreted into milk and are therefore contraindicated in nursing mothers.

ADVERSE EFFECTS

The following adverse reactions have been observed in patients receiving tetracyclines, including doxycycline.

System Organ Class	Common	Uncommon	Rare	Not known
Infections and infestations		Vaginal infection	Candida infection	
Blood and lymphatic system disorders			Haemolytic anaemia, neutropenia, thrombocytopenia, eosinophilia	
Immune system disorders	Hypersensitivity (including anaphylactic shock, anaphylactic		Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS)	Jarisch-Herxheimer reaction

	reaction, anaphylactoid reaction, angioedema, exacerbation of systemic lupus erythematosus, pericarditis, serum sickness, Henoch-Schonlein purpura, hypotension, dyspnoea, tachycardia, peripheral, oedema and urticaria)			
Endocrine disorders			Brown-black microscopic discoloration of thyroid glands	
Metabolism and nutrition disorders			Porphyria, decreased appetite	
Nervous system disorders	Headache		Anxiety, benign intracranial hypertension (pseudotumor cerebri)*, fontanelle bulging	
Ear and labyrinth disorders			Tinnitus	
Vascular disorders			Flushing	
Gastrointestinal disorders	Nausea/vomiting	Dyspepsia (Heartburn/gastritis)	Pancreatitis, pseudomembranous colitis, Clostridium difficile colitis, oesophageal ulcer, eosophagitis, enterocolitis, inflammatory lesions (with monilial overgrowth) in the anogenital region, dysphagia, abdominal pain, diarrhoea, glossitis, stomatitis	Tooth discoloration
Hepatobiliary disorders			Hepatic failure, hepatitis, hepatotoxicity, jaundice, hepatic function abnormal.	
Skin and subcutaneous tissue disorders	Photosensitivity reaction, rash including maculopapular and erythematous rashes		Toxic epidermal necrolysis, Stevens-Johnson syndrome, erythema multiform, dermatitis exfoliative, photoonycholysis	
Musculoskeletal, connective tissue and bone disorders			Arthralgia, myalgia	
Renal and urinary disorders			Blood urea increased	

* Symptoms included blurring of vision, scotomata and diplopia. Permanent visual loss has been reported.
a Reversible and superficial discolouration of permanent teeth has been reported with the use of doxycycline but frequency cannot be estimated from available data

WARNING AND PRECAUTIONS

Paediatric population

The use of drugs of the tetracycline class during tooth development (last half of pregnancy; infancy and childhood to the age of 8 years) may cause permanent discoloration of the teeth (yellow-grey-brown). This adverse reaction is more common during long-term use of the drugs but has been observed following repeated short-term courses. Enamel hypoplasia has also been reported. Use doxycycline in paediatric patients aged younger

than 8 years only when the potential benefits are expected to outweigh the risks in severe or life-threatening conditions (e.g. anthrax, Rocky Mountain spotted fever), particularly only when there are no adequate alternative therapies.

Although the risk of permanent teeth staining is rare in children aged 8 years to less than 12 years, the use of doxycycline should be carefully justified in situations where other drugs are not available, are not likely to be effective or are contraindicated.

Use in patients with impaired hepatic function

Doxycycline should be administered with caution to patients with hepatic impairment or those receiving potentially hepatotoxic drugs.

Abnormal hepatic function has been reported rarely and has been caused by both the oral and parenteral administration of tetracyclines, including doxycycline.

Use in patients with renal impairment

Excretion of doxycycline by the kidney is about 40%/72 hours in individuals with normal renal function. This percentage excretion may fall to a range as low as 1-5%/72 hours in individuals with severe renal insufficiency (creatinine clearance below 10ml/min). Studies have shown no significant difference in the serum half-life of doxycycline in individuals with normal and severely impaired renal function. Haemodialysis does not alter the serum half-life of doxycycline. The anti-anabolic action of the tetracyclines may cause an increase in blood urea. Studies to date indicate that this anti-anabolic effect does not occur with the use of Doxycycline in patients with impaired renal function.

Serious skin reactions

Serious skin reactions, such as exfoliative dermatitis, erythema multiforme, Stevens-Johnson syndrome, toxic epidermal necrolysis, and drug reaction with eosinophilia and systemic symptoms (DRESS) have been reported in patients receiving doxycycline. If serious skin reactions occur, doxycycline should be discontinued immediately and appropriate therapy should be instituted.

Photosensitivity

Photosensitivity manifested by an exaggerated sunburn reaction has been observed in some individuals taking tetracyclines, including doxycycline. Patients likely to be exposed to direct sunlight or ultraviolet light should be advised that this reaction can occur with tetracycline drugs and treatment should be discontinued at the first evidence of skin erythema. Photoonycholysis has also been reported in patients receiving doxycycline.

Benign intracranial hypertension

Bulging fontanelles in infants have been reported in individuals receiving tetracyclines. Benign intracranial hypertension (pseudotumor cerebri) has been associated with the use of tetracyclines including doxycycline. Benign intracranial hypertension (pseudotumor cerebri) is usually transient, however cases of permanent visual loss secondary to benign intracranial hypertension (pseudotumor cerebri) have been reported with tetracyclines including doxycycline. If visual disturbance occurs during treatment, prompt ophthalmologic evaluation is warranted. Since

intracranial pressure can remain elevated for weeks after drug cessation patients should be monitored until they stabilize. Concomitant use of isotretinoin or other systemic retinoids and doxycycline should be avoided because isotretinoin is also known to cause benign intracranial hypertension (pseudotumor cerebri).

Microbiological overgrowth

The use of antibiotics may occasionally result in the overgrowth of non-susceptible organisms including Candida. If aresistant organism appears, the antibiotic should be discontinued and appropriate therapy instituted.

Pseudomembranous colitis has been reported with nearly all antibacterial agents, including doxycycline, and has ranged in severity from mild to life-threatening. It is important to consider this diagnosis in patients who present with diarrhoea subsequent to the administration of antibacterial agents.

Clostridium difficile associated diarrhoea (CDAD) has been reported with use of nearly all antibacterial agents, including doxycycline, and may range in severity from mild diarrhoea to fatal colitis. Treatment with antibacterial agents alters the normal flora of the colon leading to overgrowth of *C. difficile*.

C. difficile produces toxins A and B which contribute to the development of CDAD.

Hypertoxin producing strains of *C. difficile* cause increased morbidity and mortality, as these infections can be refractory to antimicrobial therapy and may require colectomy. CDAD must be considered in all patients who present with diarrhoea following antibiotic use. Careful medical history is necessary since CDAD has been reported to occur over two month safter the administration of antibacterial agents.

Oesophagitis

Instances of oesophagitis and oesophageal ulcerations have been reported in patients receiving capsule and tablet forms of drugs in the tetracycline class, including doxycycline. Most of these patients took medications immediately before going to bed or with inadequate amounts of fluid.

Porphyria

There have been rare reports of porphyria in patients receiving tetracyclines.

Venereal disease

When treating venereal disease, where co-existent syphilis is suspected, proper diagnostic procedures including dark-field examinations should be utilised. In all such cases monthly serological tests should be made for at least four months.

Beta-haemolytic streptococci infections

Infections due to group A beta-haemolytic streptococci should be treated for at least 10 days.

Myasthenia gravis

Due to a potential for weak neuromuscular blockade, care should be taken in administering tetracyclines to patients with myasthenia gravis.

Systemic lupus erythematosus

Tetracyclines can cause exacerbation of SLE.

Methoxyflurane

Caution is advised in administering tetracyclines with methoxyflurane.

Jarisch-Herxheimer reaction

Some patients with spirochete infections may experience a Jarisch-Herxheimer reaction shortly after doxycycline treatment is started. Patients should be reassured that this is a usually self-limiting consequence of antibiotic treatment of spirochete infections.

INTERACTIONS WITH OTHER MEDICAMENTS

The absorption of doxycycline may be impaired by concurrently administered antacids containing aluminium, calcium, magnesium or other drugs containing these cations; oral zinc, iron salts or bismuth preparations. Dosages should be maximally separated.

Since bacteriostatic drugs may interfere with the bactericidal action of penicillin, it is advisable to avoid giving doxycycline in conjunction with penicillin.

There have been reports of prolonged prothrombin time in patients taking warfarin and doxycycline. Tetracyclines depress plasma prothrombin activity and reduced doses of concomitant anticoagulants may be necessary.

The serum half-life of doxycycline may be shortened when patients are concurrently receiving barbiturates, carbamazepine or phenytoin. An increase in the daily dosage of doxycycline should be considered.

Alcohol may decrease the half-life of doxycycline.

A few cases of pregnancy or breakthrough bleeding have been attributed to the concurrent use of tetracycline antibiotics with oral contraceptives.

Doxycycline may increase the plasma concentration of ciclosporin. Co-administration should only be undertaken with appropriate monitoring.

The concurrent use of tetracyclines and methoxyflurane has been reported to result in fatal renal toxicity.

Concomitant use of isotretinoin or other systemic retinoids and doxycycline should be avoided. Each of these agents used alone has been associated with benign intracranial hypertension (pseudotumor cerebri).

Laboratory test interactions

False elevations of urinary catecholamine levels may occur due to interference with the fluorescence test.

RECOMMENDED DOSAGE

It must be remembered that the usual dosage and frequency of administration of doxycycline differs from that of most other tetracyclines. Exceeding the commended dosage may result in an increased incidence of side effects. Therapy should be continued at least 24 to 48 hours after symptoms and fever have subsided. When used in

streptococcal infections, therapy should be continued for 10 days to prevent the development of rheumatic fever or glomerulonephritis.

The usual dose of doxycycline in adults is 200 mg on the first day of treatment (administered as a single dose or as 100 mg every 12 hours) followed by a maintenance dose of 100 mg/day (administered as a single dose or as 50 mg every 12 hours). In the management of more severe infections (particularly chronic infections of the urinary tract), 200 mg daily should be given throughout the treatment period.

For children above 8 years of age: The recommended dosage schedule for children weighing 45 kg or less is 4.4 mg/kg of body weight (given as a single daily dose or divided into two doses on the first day of treatment), followed by 2.2 mg/kg of body weight (given as a single daily dose or divided into two doses), on subsequent days. For more severe infections, up to 4.4 mg/kg of body weight may be used. For children over 45 kg, the usual adult dose should be used.

Tick- and louse-borne relapsing fevers and louse-borne typhus have been successfully treated with a single oral dose of 100 or 200 mg, according to severity. As an alternative to reduce the risk of persistence or relapse of tick-borne relapsing fever, doxycycline 100 mg every 12 hours for seven days is recommended.

Early Lyme disease (Stage 1): Doxycycline 100 mg orally twice daily for 14-60 days, according to clinical signs, symptoms and response.

Uncomplicated urethral, endocervical or rectal infection in adults caused by *Chlamydia trachomatis*: 100 mg, by mouth, twice daily for seven days.

Non-gonococcal urethritis (NGU) caused by *Chlamydia trachomatis* or *Ureaplasma urealyticum*: 100 mg, by mouth, twice daily for seven days.

Lymphogranuloma venereum caused by *Chlamydia trachomatis*: Doxycycline 100 mg orally twice daily for a minimum of 21 days.

Uncomplicated gonococcal infections of the cervix, rectum or urethra where gonococci remain fully sensitive: Doxycycline 100 mg by mouth twice daily for seven days plus co-treatment with an appropriate cephalosporin or quinolone is recommended, such as the following: Cefixime 400 mg orally in a single dose or ceftriaxone 125 mg intramuscularly (IM) in a single dose or ciprofloxacin 500 mg orally in a single dose or ofloxacin 400 mg orally in a single dose.

Uncomplicated gonococcal infections of the pharynx, where gonococci remain fully sensitive: Doxycycline 100 mg by mouth twice daily for seven days, plus co-treatment with an appropriate cephalosporin or quinolone is recommended, such as the following: ceftriaxone 125 mg IM in a single dose or ciprofloxacin 500 mg orally in a single dose or ofloxacin 400 mg orally in a single dose.

Primary and secondary syphilis: Non-pregnant penicillin-allergic patients who have primary or secondary syphilis can be treated with the following regimen: Doxycycline 100 mg orally twice daily for two weeks, as an alternative to penicillin therapy.

Latent and tertiary syphilis: Non-pregnant penicillin-allergic patients who have tertiary or secondary syphilis can be treated with the following regimen: Doxycycline 100 mg orally twice daily for two weeks, as an alternative to penicillin therapy if the duration of the infection is known to have been less than one year. Otherwise, doxycycline should be administered for four weeks.

Acute pelvic inflammatory disease (PID):

Inpatient: Doxycycline 100 mg every 12 hours, plus cefoxitin 2 g IV every six hours or cefotetan 2 g IV every 12 hours for at least four days and at least 24 to 48 hours after patient improves. Then continue doxycycline 100 mg by mouth twice daily to complete 14 days total therapy.

Out-Patient: Doxycycline 100 mg by mouth twice daily for 14 days as adjunctive therapy with ceftriaxone 250 mg IM once or cefoxitin 2 g IM, plus probenecid 1 g orally in a single dose concurrently once, or other parenteral third-generation cephalosporin (e.g., ceftizoxime or cefotaxime).

Acne Vulgaris: 50-100 mg daily for up to 12 weeks.

For treatment of chloroquine-resistant falciparum malaria: 200 mg daily for at least seven days. Due to the potential severity of the infection, a rapid-acting schizonticide such as quinine should always be given in conjunction with doxycycline; quinine dosage recommendations vary in different areas.

For prophylaxis of malaria: 100 mg daily in adults; for children over 8 years of age the dose is 2 mg/kg given once daily up to the adult dose.

Prophylaxis can begin 1-2 days before travel to malarious areas. It should be continued daily during travel in the malarious areas and for four weeks after the traveler leaves the malarious area.

For the treatment and selective prophylaxis of cholera in adults: 300 mg in a single dose.

For the prevention of scrub typhus: 200 mg as a single oral dose.

For the prevention of traveler's diarrhea in adults: 200 mg on the first day of travel (administered as a single dose or as 100 mg every 12 hours) followed by 100 mg daily throughout the stay in the area. Data on the use of the drug prophylactically are not available beyond 21 days.

For the prevention of Leptospirosis: 200 mg orally on a weekly basis throughout the stay in the area and 200 mg at the completion of the trip. Data on the use of the drug prophylactically are not available beyond 21 days.

For the treatment of Leptospirosis: 100 mg orally twice daily for seven days.

Published studies to date have indicated that administration of doxycycline at the usual recommended doses does not lead to excessive accumulation of the antibiotic in patients with renal impairment.

Administration

Administration of adequate amounts of fluid along with tablet form of drugs in the tetracycline class is recommended to reduce the risk of esophageal irritation and ulceration.

If gastric irritation occurs, it is recommended that doxycycline be given with food or milk. Studies indicate that the absorption of doxycycline is not markedly influenced by simultaneous ingestion of food or milk.

PREGNANCY AND LACTATION

Pregnancy

Doxycycline is contraindicated in pregnancy. It appears that the risks associated with the use of tetracyclines during pregnancy are predominantly due to effects on teeth and skeletal development.

Lactation

Tetracyclines are excreted into milk and are therefore contraindicated in nursing mothers.

OVERDOSE AND TREATMENT

Acute overdosage with antibiotics is rare. In the event of overdosage discontinue medication. Gastric lavage plus appropriate supportive treatment is indicated. Dialysis does not alter serum half-life and thus would not be of benefit in treating cases of overdosage.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

The effect of doxycycline on the ability to drive or operate heavy machinery has not been studied. There is no evidence to suggest that doxycycline may affect these abilities.

ROUTE OF ADMINISTRATION

Oral.

STORAGE CONDITIONS

Store below 30°C.

Protect from light.

SHELF LIFE

Product should not be used beyond the expiry date imprinted on the product packaging.

PRESENTATION

In blister of 500 capsules.

Date of revision: 10th April 2023

PRODUCT REGISTRATION HOLDER /MANUFACTURER:

PHARMANIAGA MANUFACTURING BERHAD (198001006232)

No.11A, Jalan P/1, Kawasan Perusahaan Bangi,

43650 Bandar Baru Bangi, Selangor Darul Ehsan, Malaysia.

PRP 0275.8 100423